

Robustness and Real-time Improvement for Vehicle Tracking by UAV based on CamShift Algorithm with Hue, Saturation, Value Components

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Abstract—This paper aims to find a new method to improve the robustness and real-time of the Continuously Adaptive Mean-Shift Algorithm(CamShift) algorithm. After the requisition of the image from the camera on the UAV, a search window is set while the RGB values of the image are transferred into Hue, Saturation, Value components(HSV components) and their values respectively as well. A 2-dimension histogram and the initial centre of the tracked vehicle are then calculated. MeanShift algorithm was coming up to decide whether it is converged. If the output is true, there will be a new (X, Y) coordinate to be considered as the new location of the search window in the next frame. If not, the MeanShift algorithm will repeat for recognition of the vehicle. This improved CamShift algorithm suggests a new method of vehicle tracking by UAV, it redresses the limitation of previous findings that complex backgrounds and enormous calculations would discount the performance of the algorithm.

Keywords: CamShift; Mean Shift; HSV components; vehicle tracking

1. Introduction

1.1. Brief View of UAV

Unmanned Aerial Vehicle (Unmanned Aerial Vehicle) is an Unmanned aircraft operated by radio remote control device and self-provided program control device, or operated completely or intermittently by on-board computer[1].

1.2. Classification

To classify it begin with the flight platform structure, UAV can be divided into fixed-wing UAV, rotor UAV, unmanned airship, umbrella wing UAV, flapping wing UAV and so on. The main research object of this paper is the rotor UAV. Rotor UAV has absolute advantages over fixed-wing UAV in medium, low altitude and low-speed operation. Also, it could take off and land vertically in the complex urban environment, meanwhile hovering in the air when necessary[2-1]. A fixed launcher and traditional runway are not required for rotor UAV, which cannot be ignored when preparing the flight of a fixed-wing UAV.

1.3. Significance

Due to its low cost and low risk, UAV has gradually infiltrated those industries that traditionally require a lot of human participation. With the development in recent years, UAV has emerged in various fields including agriculture, military, logistics, photography and so on with its unique characteristics. To be more specific, UAV tracking is one of the most famous fields currently. Contributed to the remarkable achievements of the traditional monitoring of cars by surveillance cameras, the police could utilize them to track any suspicious vehicles that threaten the city's safe, catch the suspect and provide a peaceful place for the civilians. Having said that, the irremovable characteristic makes it hard to accomplish the purpose of all-condition tracking. Thus, the idea of introducing UAV to detect and track vehicles are promoted. Rotor UAV has a comparatively noise-free structure[2-2] and the concealment makes it possible to track the target with a reasonable distance at relatively low altitudes. In today's complex urban environment and military purpose, rotor UAV can complete real-time monitoring and positioning tracking of targets without the need for numerous fixed

surveillance cameras on the streets. To achieve the need for UAV tracking, a series of the algorithm are created. One of the most famous algorithms is the MeanShift algorithm. However, it has some problems such as easy to lose target which is moving rapidly and failing to track when the size of the object is varying all the time. To adapt to the demand for UAV tracking systems, a brand-new CamShift algorithm[3] based on the Hue component occurred. This was great progress in the UAV tracking field and helped to finish lots of academic studies. However, there are still some drawbacks that need to be overcome such as slow response[4-1] and interference from the background of the image[4-2]. Research groups in these years have tried to analyze some efficient methods for UAV tracking.

1.4. Research Status

1.4.1. Domestic Status

Li Zhang[5] proposed a method of shadow detection. Transferring RGB to HSV space of the image, and comparing the current frame with the established model. Then, decide whether the pixel is the background or object, and divide the shadow from the frame. The object selected by this process can be used for further tracking. The difficulty is that shadow detection may increase the processing time. Ruihua Dong[6] designed a vehicle tracking system by applying GPS and GIS. However, its real application is limited by signal conditions and may not work well when losses signal.

1.4.2. Worldwide Status

Margrit Betke suggested a hard real-time operating system called “Maruti”[7] to accomplish real-time multiple vehicle detection and tracking. His findings reduced the amount of calculation that the computer spent and had a quite low cost on software. Heintz[8] 's team used an implementation system for achieving a high level of situational awareness of traffic conditions in urban areas. It uses color images and thermal images as input sequences to construct and maintain qualitative object structure, recognizing the traffic behavior of tracked vehicles in real-time [9] with remarkable effect.

1.5. Analysis

In most of the cases, researchers mainly focused on shape-based and color-based tracking methods. For a simple background like the river, for example, the color-based method is comparatively effective and not likely to lose the target. On the other hand, a shape-prefer method will be more adaptive in a complex urban situation which has multiple color features[10-1]. But the size of the tracked object like vehicles changes as the position that UAV observes them is distinguishing.

There might not be a single good enough method that satisfies all the conditions.

In the case of tracking vehicles, the RGB value is mostly constant despite some tiny changes under different weather conditions. This paper is going to combine color segmentation and machine learning based on loops that participated by CamShift, MeanShift algorithms and search window to reduce the interference effect of the background, improve robustness and achieve real-time for complex background.

2. CamShift Algorithm

CamShift algorithm is an improved method of MeanShift algorithm by introducing the use of color features mainly focused on the Hue value. By setting a Region of Interest(ROI) and search window, the initial size and centre of the object can be calculated by CamShift[11-1]. Then, MeanShift will get the centre of the color distribution probability histogram. The new location of the object in the next frame will be calculated and it will help to get the new location of the search window for the next frame. These steps are repeated if the object still exists, or it will go back to recognize the object[10-2].

3. Hue, Saturation, Value Components

Color models, such as HSV and RGB, are widely used as a way to create Web graphics[12-1]. If we keep zooming in on the image that appears on the mobile phone, for example, a series of composited dots with red, green, and blue can be seen. These three colors are known as the primary colors. Hue is the description of color part of the three primary colors, and is often pictured using a cone or a cylinder[12-2].

Different Hue consists of distinguished degrees of the cylinder[13-1] as shown below in Fig. 1.

Red falls between 0 and 60 degrees.
Yellow falls between 61 and 120 degrees.
Green falls between 121 and 180 degrees.
Cyan falls between 181 and 240 degrees.
Blue falls between 241 and 300 degrees.
Magenta falls between 301 and 360 degrees.

Figure 1 Hue Value

Saturation refers to the intensity of color in an image, or how much is the amount of gray or shadow in the image, from 0 to 100%. The third component, value, represents the brightness of the picture, from 0 to 100%, where 0 is fully black [13-2].

4. Improved CamShift Algorithm Based on HSV Components

4.1. Algorithms

4.1.1. Previous Short-comes

In the past, CamShift only use the Hue value in the processing of tracking objects. The difficulty is that different objects may have the same Hue value. In this paper, the main target is vehicles on urban streets. In the urban area, like a crossroad, the color feature of the background is always varying, which makes locating vehicles very hard if only applying Hue value. Thus, in this work, an improved CamShift algorithm based on HSV components is promoted.

The algorithm[11-2] to convert RGB into HSV is shown below. H is the Hue value, S is color saturation, and V is the luminescent value. In this case, r, g, b indicates red, green, and blue components.

$$H = \begin{cases} \frac{\pi}{3} \left(\frac{b-r}{\max(r,g,b)-\min(r,g,b)} \right) & \text{if } g = \max(r,g,b) \\ \frac{\pi}{3} \left(2 + \frac{b-r}{\max(r,g,b)-\min(r,g,b)} \right) & \text{if } b = \max(r,g,b) \\ \frac{\pi}{3} \left(4 + \frac{b-r}{\max(r,g,b)-\min(r,g,b)} \right) & \text{if } r = \max(r,g,b) \end{cases} \quad (1)$$

$$S = \frac{\max(r,g,b)-\min(r,g,b)}{\max(r,g,b)} \quad (2)$$

$$V = \max(r,g,b) \quad (3)$$

The model of the HSV components is demonstrated on the Fig. 2.

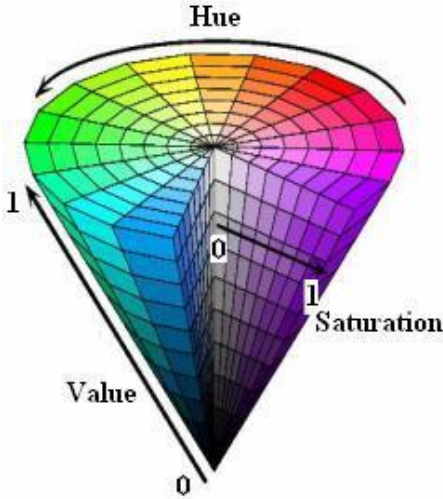


Figure 2 HSV model

In an image taken by the UAV-mounted camera on a rotor UAV, there might be a lot of distractions to affect the reliability of CamShift algorithm. When the target vehicle is confirmed by initially setting the search window. Processing of the whole image can be avoided and just focuses on the small region of the search window. This would be beneficial, as a result, reducing the processing time and approaching real-time performance[10-3].

4.2. Principle of the Camshift Algorithm Based on HSV Components

On the urban streets, the background is unpredictable as there are not only different vehicles but also various buildings and walls, for example. It requires a very large amount of calculation to consider the color information of the meaningless background besides the target. Therefore, processing before tracking is used in this work to improve accuracy and reduce reaction time.

Meanwhile, images are taken at several frames per second depending on the performance of the camera. Every single centre and size of the target is formed by the Camshift algorithm which avoids fails of locating and tracking. To achieve real-time performance, a search window would be helpful to initially set a region of interest, ignoring the rest of the image taken by the mounted camera on the UAV.

For the vehicles on the streets, only applying Hue components is not reliable enough because there are distractions that affect the Hue value of the target vehicle. As a result, HSV components are considered to be useful for a more complex environment. By creating a color segmentation possibility histogram of the vehicle, the new centre and position vector can be analyzed in the following frames. The main principle of this improved algorithm is shown in the steps below, with a flowchart in Fig. 3.

- (1) Set the search window[10-4][14] and obtain HSV values by converting from RGB values of the image.
- (2) The 2-D histogram[15] can be calculated and help to find the initial centre and position of the object, in this case is vehicles.
- (3) The new centre and position can be found by applying the MeanShift algorithm.
- (4) Repeat step 2 if the target still exists, otherwise being ready for a new round of recognition.

The centre and position of the color probability distribution can be obtained by the following equations.

$$X_c = \frac{M_{10}}{M_{00}}, y_c = \frac{M_{01}}{M_{00}} \quad (4)$$

where,

$$M_{00} = \sum_x \sum_y I(x,y) \quad (5)$$

$$M_{10} = \sum_x \sum_y xI(x,y) \quad (6)$$

$$M_{01} = \sum_x \sum_y yI(x,y) \quad (7)$$

$I(x,y)$ is the color probability value of the point (x,y)

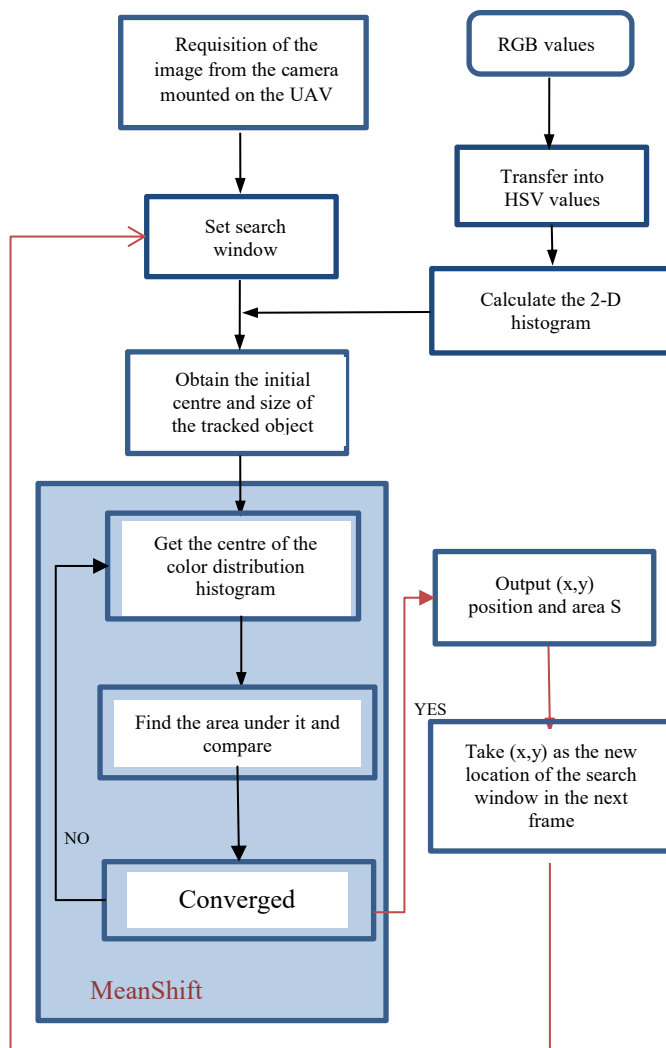


Figure 3 Improved CamShift Algorithm Flowchart

5. Conclusion

This paper suggests a possible method to replace the traditional CamShift algorithm which based on Hue only. Contribute to the color segmentation of moving vehicles, a 2-D color possibility distribution histogram plays an important role in the pre-processing of this algorithm. The use of search window also provides real-time performance. Compare to the CamShift algorithm in the past, color-based method for vehicle tracking may satisfy more situations than shape-based method. However, related experiments are not enough to prove it. Also, if there are some obstructs like multiple car paint or color changing paint, due to different temperatures, on a single vehicle, this method may become unreliable. In this case, it requires the later related and supportable data and evidence to make it convincing.

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